

CONCEPT DESIGN PRESENTATION

Luxury Equestrian Stable

A Validated, Owner-Confirmed Concept Redesign

Site	40 × 50 m (2,000 m²)
Program	40 scheduled spaces
Scheduled area	1,648.01 m² (82.4% coverage)
Validation	PASS — 27 geometric checks, 0 failed

Concept plan only. Not for construction or permit use until reviewed and approved by a licensed local architect/engineer, MEP/civil engineer, and fire/life-safety consultant.

Prepared as a concept design package. See Section 16 for a full list of items requiring licensed professional review before any construction.

Table of Contents

01	Executive Summary
02	Project Goals
03	Site Summary
04	Design Concept & Philosophy
05	Zoning Strategy
06	Master Plan
07	Dimensions & Design Logic
08	Zone Narratives
09	Circulation & Operations
10	Materials & Finishes Palette
11	Render Gallery
12	Technical Highlights
13	Room Schedule Summary
14	Advantages Over the Original Scheme
15	Next Steps
16	What Still Requires Licensed Professional Review
17	Appendix & Reference Index

SECTION 01

Executive Summary

This package presents a fully validated concept redesign for a private luxury equestrian stable on a 40.00 × 50.00 m (2,000 m²) site. The design delivers the complete owner brief — 20 standard stalls, 2 premium stalls, 2 veterinary/isolation rooms, feed and service rooms, two equal paddocks, full worker accommodation, a men's majlis and outdoor majlis, a private bedroom suite, and a parking apron — organized into a clear front-to-back zoning strategy that separates guest arrival, the working stable, and the paddock/service zone.

Every dimension in this package is mathematically validated: 27 geometric checks pass with zero failures, both horizontal and vertical dimension chains close exactly to the site boundary, and total scheduled area is 1,648.01 m² — a coverage ratio that leaves deliberate landscaped and circulation space rather than crowding the site.

This is a concept design package, not a construction document. Section 16 states plainly what still requires licensed structural, MEP, fire/life-safety, and civil review before any permitting or construction step.

Project Goals

The redesign was directed by five goals, each carried through into the final validated layout:

- **Deliver the complete program** — every required room from the owner's brief and CAD reference is present at its confirmed size, with nothing invented or dropped.
- **Operate efficiently** — horse care, feeding, and service routes should be as short and uncrossed as possible.
- **Present as a luxury estate** — the arrival and guest sequence should feel generous and considered, not utilitarian.
- **Keep paddocks practical** — both paddocks equal in size and directly serviceable, not reduced to leftover land.
- **Stay honest** — every dimension traceable to a source (the written brief, the CAD drawing, or an explicit owner decision), with no silent assumptions.

SECTION 03

Site Summary

The site measures 40.00 m wide by 50.00 m deep (2,000 m² total), with the origin at the southwest corner and Y=0 confirmed as the entrance/gate edge from the original hand sketch. The design uses 1,648.01 m² (82.4%) for the 40 scheduled spaces; the remaining area is landscaped buffer, transition zones, and aisle circulation, left deliberately open for a calm, uncluttered feel rather than maximized for buildable area.

SECTION 04

Design Concept & Philosophy

The design language is calm and materials-first: proportion and shade rhythm carry the luxury feel rather than ornament.

- Twin symmetric courtyards (indoor majlis + outdoor majlis) create the arrival "moment" without visual clutter.
- A single continuous shade canopy over the stable aisle unifies all 24 horse rooms visually and functionally.
- One consistent roofline covers the entire service/worker/feed wing rather than three disconnected sheds.
- Equal, generously proportioned paddocks read as a designed landscape feature, not leftover land.

SECTION 05

Zoning Strategy

The site is organized into nine bands along its depth (front to back), each closing exactly against the 40.00 × 50.00 m boundary:

- **Motor court + parking** — guest arrival, drop-off, four-bay parking apron.
- **Guest/owner band** — majlis and outdoor majlis mirrored either side of a landscaped view corridor; the private bedroom suite tucked to the east, private from the majlis.
- **Privacy/transition buffer** — landscaped, controls the sightline from guest to working stable.
- **Stall Row A / central aisle / Stall Row B** — the 20 standard stalls in a double-loaded barn, aisle-facing, non-slip flooring, shaded canopy above.
- **Feed / service / premium / veterinary / worker wing** — one unified roofline holding everything the stable needs day-to-day, one aisle from the horses, one wall from the paddocks.
- **Vestibule** — a gated transition from the working wing to the paddock spine.
- **Paddocks + wash/manure spine** — two equal paddocks flanking a dedicated service yard.

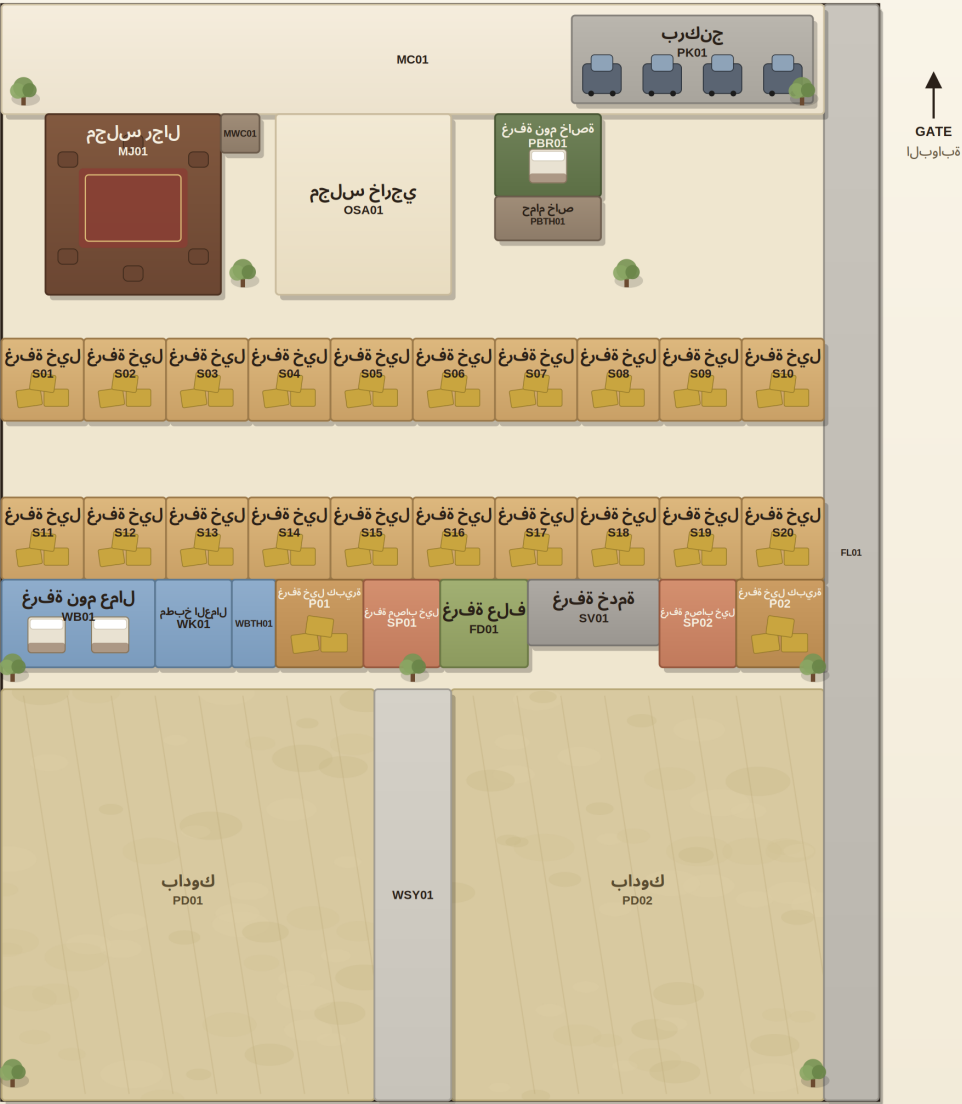
A continuous 2.50 m perimeter fire/service lane runs the full site depth along the east edge, connecting the front gate straight through to the rear paddock spine without ever crossing the guest motor court or the majlis.

Master Plan

LUXURY EQUESTRIAN STABLE

Premium Concept Masterplan — Finalized Validated Layout

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LEGEND

- | | |
|---|-----------------------------------|
| Standard Horse Stall — 3.75×3.75 m (×20) | Men's Majlis — 8.00×8.24 m |
| Premium Horse Stall — 4.00×4.00 m (×2) | Private Bedroom Suite |
| Veterinary / Isolation Room ('mesab', owner-confirmed) — 3.50×4.00 m (×2) | Outdoor Majlis / Sitting Terrace |
| Feed Room — 4.00×4.00 m | Paddock — 2× equal, 318.9 m² each |
| Service Room — 6.00×3.00 m (owner-confirmed) | Parking Apron — 4 bays |
| Worker Accommodation | |

Concept plan only. Not for construction or permit use until reviewed and approved by a licensed local architect/engineer, MEP/civil engineer, and fire/life-safety consultant.

Premium vector cutaway master plan — generated directly from the validated coordinate model. Every rectangle shown is the exact, checked geometry; nothing here is illustrative guesswork.

SECTION 07

Dimensions & Design Logic

Horizontal (X) chain — 40.00 m exactly:

Program width (stalls, wing, guest rooms, paddocks): 0 → 37.5 m, plus the perimeter service / emergency fire lane: 37.5 → 40 m.

Vertical (Y) chain — 50.00 m exactly:

Motor court/parking (5.00 m) + guest rooms band (8.24 m) + landscaped transition buffer (2.00 m) + Stable Row A (3.75 m) + central aisle (3.50 m) + Stable Row B (3.75 m) + feed/service/premium/veterinary/worker wing (4.00 m) + vestibule (1.00 m) + paddocks and service spine (18.76 m) = 50.00 m.

Both chains are re-verified automatically every time the design is regenerated — they are not measured once and assumed to still hold; validate.py confirms both totals every run.

Zone Narratives

The Equestrian Zone

Twenty standard 3.75×3.75 m stalls and two premium 4.00×4.00 m stalls share one continuous, shaded central aisle rather than being split across separate wings — every stall opens directly onto shared circulation, shortening every staff round. The two veterinary/isolation rooms sit at the aisle's southern end, immediately adjacent to feed and service, naturally separated from the general stall population without requiring a detour.

The Guest & Majlis Zone

Arrival flows from the motor court into a pair of symmetric courtyards — the indoor men's majlis and the outdoor majlis/sitting terrace — before reaching the private bedroom suite. A deliberate landscaped buffer stands between this zone and the working stable, so the estate's most prominent, generous frontage is never interrupted by stable traffic.

The Worker & Service Wing

Worker bedroom, kitchen, and bathroom share one wing and one roofline with the feed room, service room, and both veterinary/isolation rooms — instead of being buried in a narrow slot between stall columns. Feed and service sit at the center of the wing, equidistant from both stall rows, minimizing walking distance for daily routines.

The Private Suite & Paddocks

The private bedroom and bathroom sit east of the majlis, private from the guest sequence. To the rear, two equal 318.92 m² paddocks flank a dedicated wash/manure service spine that connects directly to both paddock gates and the perimeter fire lane — turnout and stable maintenance never cross guest or general circulation.

Circulation & Operations

- **Horse circulation:** every stall opens directly onto the single shared central aisle; no horse ever needs to cross the guest zone or motor court.
- **Service/feed circulation:** feed and service rooms sit centered in the wing, equidistant from both stall rows; the perimeter fire/service lane lets a feed truck or farrier reach the wing or paddocks without touching the guest motor court.
- **Guest circulation:** arrival → motor court → majlis or outdoor majlis → private suite, with a deliberate visual and physical stop before the working stable begins.
- **Manure/wash logistics:** the dedicated wash/manure service spine connects directly to both paddock gates and the perimeter fire lane, so soiled bedding never crosses the stable aisle or guest zone.
- **No dead corners or bottlenecks:** the aisle is a uniform 3.50 m throughout, the perimeter lane a uniform 2.50 m, and every enclosed room has exactly one described door or gate.

Materials & Finishes Palette

Element	Material Direction
Stall / wing walls	Warm sandstone/limestone render, bronze accent trim on premium stalls
Stable doors	Solid warm walnut timber Dutch doors, powder-coated steel yoke bars
Majlis / private suite walls	Deep espresso plaster (majlis), olive-toned plaster (private suite)
Roofs	Bronze standing-seam metal, consistent across all structures
Paddock fencing	Dark walnut-toned timber post-and-rail, two rail bands
Terrace / majlis flooring	Warm limestone paving
Motor court / fire lane	Limestone paving (guest) vs. utilitarian grey paving (service)
Landscape	Native drought-tolerant planting, shade trees at guest and paddock-edge nodes
Lighting concept	Warm 2700K low-glare bollards along aisle and lane; uplighting at majlis facade only

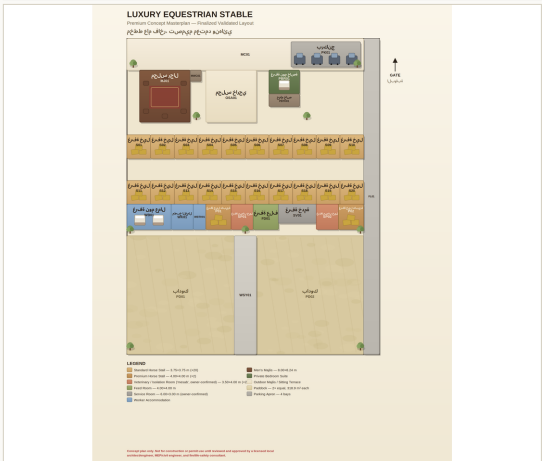
SECTION 11

Render Gallery

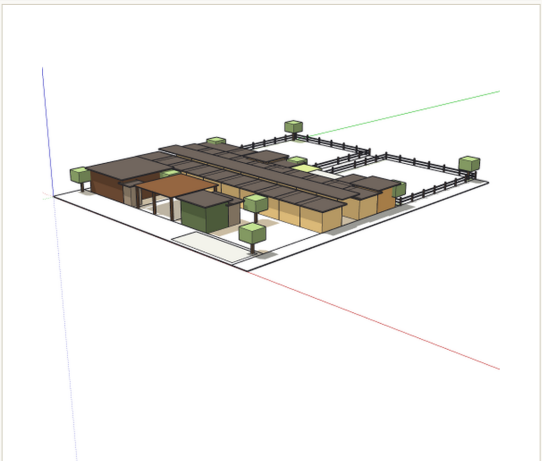
The primary visualization for this package is a premium vector cutaway illustration, generated deterministically from the validated coordinate model — every wall, fill, and label traces back to an exact, checked dimension, eliminating any risk of a rendered image misrepresenting the actual design. A supporting 3D massing model (geometrically synchronized with this same validated layout) is also included. See RENDER_PRODUCTION_NOTES.md for a full, honest account of fidelity and tooling.

Luxury Equestrian Stable — Render Contact Sheet

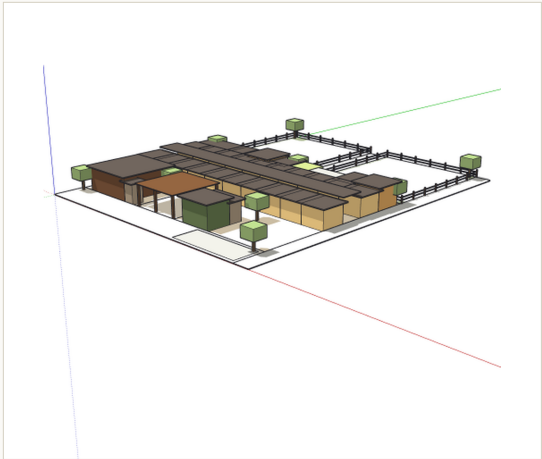
Concept visualization package — not for construction or permit use



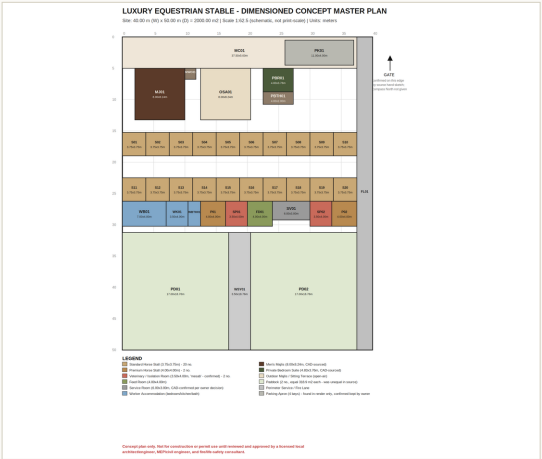
1. Premium Vector Cutaway Master Plan (primary deliverable)



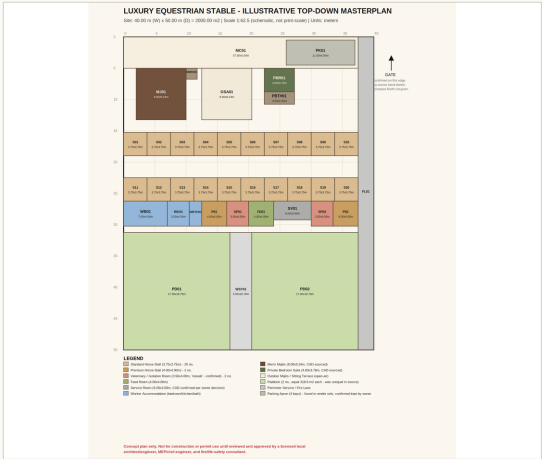
2. 3D Aerial Massing - Synchronized Geometry (SketchUp, prior pass)



3. 3D Massing Thumbnail - Synchronized Geometry (SketchUp, prior pass)



4. Dimensioned Technical Plan



5. Styled Masterplan (SVG render)

Item 1 is generated directly, deterministically from the validated coordinate model (design/scripts/model.py) - zero hallucination risk.
Items 2-3 reflect the finalized, owner-confirmed geometry (synchronized in the prior 3D-sync pass); see RENDER_PRODUCTION_NOTES.md for full disclosure.

Contact sheet of all current visual deliverables.

Technical Highlights

Geometric checks passed	27
Geometric checks failed	0
Disclosed warnings	1 (documented trade-off, not a defect)
Owner-confirmed decisions	5
X dimension chain	40.00 m (target 40.00 m)
Y dimension chain	50.00 m (target 50.00 m)
Paddock area consistency	Both paddocks equal at 318.92 m² each

Every figure above is computed live from design/scripts/model.py and validation_report.json each time the package is regenerated — none of it is hand-typed, so it cannot silently drift out of sync with the actual design.

SECTION 13

Room Schedule Summary

Function	Count	Area (m²)	% of Scheduled Area
Paddocks	2	637.84	38.7%
Horse Stalls (Standard)	20	281.25	17.1%
Arrival / Motor Court	1	187.50	11.4%
Fire Lane / Service Lane	1	125.00	7.6%
Men's Majlis	1	65.92	4.0%
Outdoor Majlis / Sitting	1	65.92	4.0%
Service Yard	1	65.66	4.0%
Parking Apron	1	44.00	2.7%
Horse Stalls (Premium)	2	32.00	1.9%
Worker Bedroom	1	28.00	1.7%
Veterinary / Isolation Rooms	2	28.00	1.7%
Private Bedroom	1	18.16	1.1%
Service Room	1	18.00	1.1%
Feed Room	1	16.00	1.0%
Worker Kitchen	1	14.00	0.8%
Private Bathroom	1	9.66	0.6%
Worker Bathroom	1	8.00	0.5%
Men's WC	1	3.10	0.2%
TOTAL	40	1648.01	100.0%

Full 40-space schedule with individual coordinates is available in the Technical Appendix (Sheet 04) and in [design/output/schedules/room_schedule.csv](#).

SECTION 14

Advantages Over the Original Scheme

- **Horse zone logic:** the same 20+2 stall count and sizes, but the two rows now share one continuous shaded aisle rather than being split across separate wings, shortening every staff round.
- **Paddock placement:** both paddocks are now equal and directly reachable from the working wing via a dedicated wash/manure spine — turnout no longer requires crossing guest or general circulation.
- **Guest/majlis sequence:** arrival → motor court → majlis/outdoor majlis → private suite, with a clear visual and physical stop before the stable begins.
- **Worker/service separation:** worker housing and feed/service now share one wing and one roofline instead of being buried in a narrow slot between stall columns.
- **Feed/service efficiency:** feed and service rooms sit at the center of the wing, equidistant from both stall rows.
- **Visual hierarchy:** guest wing at the most prominent frontage, stable in the calm productive middle, service/paddocks at the working rear — a legible luxury-estate hierarchy.

Next Steps

- Engage a licensed local architect/engineer to convert this concept into permit-ready construction documents.
- Commission structural, MEP, fire/life-safety, and civil/drainage consultants (see Section 16).
- Confirm local zoning, agricultural/equestrian-use, and building code requirements before any site work.
- Use the CAD underlay (stable_concept_underlay.dxf) as a coordinated starting reference for the professional design team — it is explicitly marked as a concept underlay, not a construction drawing.
- Review material and finish selections with the owner once a contractor/quantity surveyor is engaged, since this package specifies material direction, not final specifications.

SECTION 16

What Still Requires Licensed Professional Review

This package is a concept design only. The following items are explicitly not resolved and require a licensed professional before any construction or permitting step:

- Structural design, footings, and roof framing — this is a massing/concept model only.
- MEP: electrical, plumbing, ventilation/HVAC sizing for all enclosed rooms, especially the worker kitchen/bathroom and the veterinary function assigned to SP01/SP02.
- Fire/life-safety review of egress widths, the perimeter fire lane's actual vehicle turning radius, and manure/hay fire separation distances.
- Civil/drainage design for the paddocks, wash-down spine, and site-wide stormwater management.
- Local building code, zoning, and agricultural/equestrian-use permitting compliance — none of which has been verified in this exercise.
- The veterinary/isolation fit-out for SP01/SP02 (ventilation, drainage, exam lighting) still needs a licensed veterinary/MEP consultant, even though the room function itself is confirmed.

Concept plan only. Not for construction or permit use until reviewed and approved by a licensed local architect/engineer, MEP/civil engineer, and fire/life-safety consultant.

SECTION 17

Appendix & Reference Index

This presentation draws on, and can be cross-checked against, the following source files in the project repository:

<code>design/scripts/model.py</code>	Single coordinate source of truth for all 40 spaces
<code>design/scripts/validate.py</code>	Geometric validation logic (overlap, boundary, dimension checks)
<code>design/output/reports/source_audit.md</code>	Full technical audit and source-conflict record
<code>design/output/reports/technical_appendix.pdf</code>	Cover, metrics, master plan, and full room schedule sheets
<code>design/output/schedules/room_schedule.csv</code>	Full 40-space schedule, machine-readable
<code>design/output/schedules/coordinate_schedule.csv</code>	Full coordinate schedule, machine-readable
<code>design/output/plans/premium_cutaway_masterplan.svg / .png</code>	Primary vector illustration
<code>design/output/plans/plan_dimensioned.svg / .png</code>	Dimensioned technical plan
<code>design/output/plans/stable_concept_underlay.dxf</code>	CAD underlay — concept only, not for construction
<code>RENDER_PRODUCTION_NOTES.md</code>	Full disclosure of render tooling and fidelity
<code>SOURCE_OF_TRUTH_LOCK.md</code>	Frozen geometry, program, and owner-decision reference

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